Prime Quintuplets and the Sextuplet Boundary for $Q(n) = n^{47} - (n-1)^{47}$

Extreme Prime Clusters in a High-Degree Polynomial

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Abstract

Among the 742 prime quadruplets discovered for $Q(n) = n^{47} - (n-1)^{47}$ over $1 \le n \le 2\times10^{11}$ [4], exactly **7 extend to prime quintuplets** (five consecutive integers all generating probable primes of 487–519 digits), while **no sextuplet** was found. This paper develops a quantitative theory for this emergence–extinction transition.

We prove that $Q(n)$ has a *bifurcated* root structure: $\omega_1(p) = 0$ for all primes $p \not\equiv 1 \pmod{47}$, but $\omega_1(p) = 46$ for all $p \equiv 1 \pmod{47}$. The primes in the latter class—which we call *resonant primes*—impose a *periodic sieve obstruction* that produces dramatic downward corrections ("cliffs") in the Bateman–Horn Euler product, the first occurring at $p = 283$. Computing the singular series to sieve bound $B = 10{,}000$ with the correct root counts, and calibrating against the empirical quadruplet count, yields $\mathfrak{S}_5 \approx 57{,}100$ and $\mathfrak{S}_6 \approx 520{,}000$. The resulting predictions—$E[C_5(2\times10^{11})] = 5.83$ and $E[C_6(2\times10^{11})] = 0.047$—agree with observation (7 quintuplets, 0 sextuplets) to within Poisson noise. The first sextuplet is predicted at $\boldsymbol{N^* \approx 1.05 \times 10^{13}}$, requiring a 52-fold extension of the current search.

Keywords: prime quintuplets, sextuplet boundary, Bateman–Horn conjecture, singular series, periodic sieve obstruction, resonant primes, constellation hierarchy, high-degree polynomials

MSC 2020: 11N32, 11A41, 11Y11, 11Y16

1 Introduction

A *prime k-constellation* of the polynomial $Q(n) = n^{47} - (n-1)^{47}$ is a run of k consecutive integers $n, n+1, \ldots, n+k-1$ such that all k values of Q are (probable) primes. The foundational census of the pioneer zone ($n \leq 2 \times 10^9$) [3] established a strict morphological hierarchy—18.1 million solitary primes, 173,351 pairs, 1,749 triplets, 15 quadruplets—with inter-level suppression ratios of approximately 100×. The deep-space census [4] extended the quadruplet search to $N = 2 \times 10^{11}$, finding 742 quadruplets, 7 quintuplets, and zero sextuplets.

The present paper (Part III of the Titan Project) addresses the central question arising from these data:

> *Why do exactly 7 quintuplets emerge from 742 quadruplets, and why does the sextuplet order remain vacant across 200 billion integers?*

We develop a complete quantitative answer through:

(i) A proof that $Q(n)$ possesses a *bifurcated root structure*: zero roots for most primes, but exactly 46 roots modulo every prime $p \equiv 1 \pmod{47}$ (the *resonant primes*);

(ii) Explicit computation of the singular series $\mathfrak{S}_4, \mathfrak{S}_5, \mathfrak{S}_6$ via Euler products to sieve bound $B = 10{,}000$, incorporating the correct root counts $\omega_k(p)$ at each resonant prime;

(iii) A conditional extension model predicting quintuplet counts from the quadruplet catalog with no free parameters;

(iv) A quantitative prediction for the sextuplet boundary at $N^* \approx 1.05 \times 10^{13}$;

(v) Analysis of the spatial distribution and modular structure of the 7 quintuplets.

1.1 Context: The Emergence–Extinction Transition

The progression from quadruplets ($k = 4$) to quintuplets ($k = 5$) to sextuplets ($k = 6$) represents a phase transition in the constellation hierarchy. At each level, the expected count drops by a factor of approximately 127, reflecting the interplay between the singular series ratio $\mathfrak{S}_{k+1}/\mathfrak{S}_k$ and the logarithmic integral ratio I_{k+1}/I_k. When this factor pushes the expected count below 1, the constellation order "goes extinct" in the observed range. The *sextuplet boundary* is the critical N^* at which $E[C_6(N^*)] = 1$.

2 The Bifurcated Root Structure of $Q(n)$

2.1 Main Theorem

Theorem 2.1 (Bifurcated Root Structure). *Let p be a prime and let $Q(n) = n^{47} - (n-1)^{47}$. Then:*

(a) If $p \nmid n(n-1)$, then $Q(n) \equiv 0 \pmod{p}$ if and only if $(n/(n-1))^{47} \equiv 1 \pmod{p}$, i.e., $n/(n-1)$ is a nontrivial 47th root of unity modulo p. Such roots exist if and only if $47 \mid (p-1)$, i.e., $p \equiv 1 \pmod{47}$.

(b) If $p \mid n$, then $Q(n) \equiv 1 \pmod{p}$. If $p \mid (n-1)$, then $Q(n) \equiv 1 \pmod{p}$. Neither boundary case produces a root.

(c) ***Root count:***

$$\omega_1(p) = \begin{cases} 46 & \text{if } p \equiv 1 \pmod{47}, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

Proof. **Part (a).** Suppose $p \nmid n$ and $p \nmid (n-1)$. Write $a = n$, $b = n-1$, so $a - b = 1$. Then $Q(n) \equiv 0$ iff $a^{47} \equiv b^{47}$, iff $(a/b)^{47} \equiv 1 \pmod{p}$. Let $g = \gcd(47, p-1)$. Since 47 is prime, either $g = 1$ or $g = 47$.

If $g = 1$: the only solution of $x^{47} \equiv 1$ in $\mathbb{F}_p^*$ is $x = 1$, so $a/b \equiv 1$, giving $a \equiv b$, hence $1 = a - b \equiv 0 \pmod{p}$—impossible. Thus $\omega_1(p) = 0$.

If $g = 47$: then $p \equiv 1 \pmod{47}$ and $\mathbb{F}_p^*$ contains a cyclic subgroup of order 47, generated by $\zeta = g_0^{(p-1)/47}$ where g_0 is a primitive root mod p. The equation $(a/b)^{47} = 1$ has 47 solutions $a/b = \zeta^j$ for $0 \le j < 47$. The case $j = 0$ gives $a = b$, excluded as above. For each $j \in \{1, \ldots, 46\}$: $a = b + 1$ yields $b(\zeta^j - 1) = 1$, so $b = (\zeta^j - 1)^{-1}$ (well-defined since $\zeta^j \neq 1$), giving the unique root $n = b + 1$. We verify $p \nmid b$ (since b is a unit) and $p \nmid a$ (since $a = \zeta^j b \neq 0$).

Part (b). If $p \mid n$, then $Q(n) \equiv 0 - (-1)^{47} = 1 \pmod{p}$. If $p \mid (n-1)$, then $n \equiv 1$ so $Q(n) \equiv 1 - 0 = 1 \pmod{p}$. □

Explicit example. The smallest resonant prime is $p = 283$ $(= 6 \times 47 + 1)$. A primitive 47th root of unity is $\zeta = 2^{(283-1)/47} = 2^6 = 64$, since $64^{47} \equiv 1 \pmod{283}$. Taking $j = 1$: $b = (\zeta - 1)^{-1} = 63^{-1} \equiv 9 \pmod{283}$ (since $63 \times 9 = 567 = 2 \times 283 + 1$). Thus $n = 10$, and indeed $Q(10) = 10^{47} - 9^{47} \equiv 0 \pmod{283}$.

2.2 Resonant Primes and Periodic Sieve Obstruction

Definition 2.2 (Resonant Primes). *A prime p is* resonant *for $Q(n) = n^{47} - (n-1)^{47}$ if $p \equiv 1 \pmod{47}$. The set of resonant primes is $\mathcal{R} = \{283, 659, 941, 1129, 1223, 1693, \ldots\}$. By Dirichlet's theorem, $\mathcal{R}$ is infinite; among primes up to N, approximately $1/\varphi(47) = 1/46$ of them are resonant.*

The bifurcated root structure creates a *periodic sieve obstruction*: Q passes freely through most primes ($\omega_1 = 0$) but encounters a dense 46-root barrier at every resonant prime. Each resonant prime p produces a *cliff* in the Euler product (Figure 1), multiplying the running singular series by a factor substantially less than 1.

For k-constellations, the obstruction count $\omega_k(p)$—the number of residues r where at least one of $Q(r), \ldots, Q(r+k-1)$ vanishes mod p—exceeds ω_1 due to the k shifted polynomials. Table 1 gives the computed values at the first five resonant primes.

Table 1: Root counts $\omega_k(p)$ at the first five resonant primes, and the resulting local Bateman–Horn factors $\sigma_k(p) = (1 - \omega_k/p)/(1 - 1/p)^k$.

p	ω_1	ω_4	ω_5	ω_6	$\sigma_4(p)$	$\sigma_5(p)$
283	46	126	148	164	0.563	0.486
659	46	160	194	228	0.762	0.714
941	46	176	216	256	0.816	0.778
1129	46	166	202	235	0.856	0.820
1223	46	162	195	226	0.870	0.841
Non-resonant primes (e.g. $p = 2, 3, 5, \ldots$)					$(p/(p-1))^4$	$(p/(p-1))^5$

The first resonant prime $p = 283$ is the most destructive: $\sigma_4(283) = 0.563$, meaning it *halves* the running Euler product for quadruplets. Subsequent resonant primes produce progressively milder corrections as $\omega_k/p \to 0$.

2.3 Comparison with a Root-Free Polynomial

If $\omega_k(p) = 0$ for all p, the singular series would be $\mathfrak{S}_k^{\text{free}}(10{,}000) = 8{,}604$ for $k = 4$, about 35% higher than the correct $\mathfrak{S}_4 = 6{,}392$. The 28 resonant primes below 10,000 account for this deficit,

with $p = 283$ alone responsible for a 44% reduction. Resonant primes, though sparse ($\sim 1/46$ of all primes), have a *disproportionate impact* on the singular series due to their exceptionally large root counts.

3 Computing the Singular Series

3.1 Correct Euler Products

The Bateman–Horn singular series for k-constellations is

$$\mathfrak{S}_k = \prod_{p \text{ prime}} \sigma_k(p), \qquad \sigma_k(p) = \frac{1 - \omega_k(p)/p}{(1 - 1/p)^k}. \tag{2}$$

We compute $\omega_k(p)$ by exhaustive enumeration for every prime $p \leq 10{,}000$ and form the partial product. Table 2 reports the convergence.

Table 2: Partial singular series $\mathfrak{S}_k(B)$ with correct root counts. The "root-free" column assumes $\omega_k = 0$.

Sieve bound	$\mathfrak{S}_4(B)$	$\mathfrak{S}_5(B)$	$\mathfrak{S}_6(B)$	$\mathfrak{S}_4^{\text{free}}(B)$
$B = 100$	3,389	27,350	221,400	3,389
$B = 283^-$	3,601	30,840	268,800	3,601
$B = 283$	**2,026**	**14,980**	**115,500**	3,614
$B = 1000$	3,817	34,274	313,870	8,604
$B = 5000$	5,694	51,345	471,222	—
$B = 10000$	6,392	57,173	520,348	—
Calibrated	**6,385**	**57,108**	**519,756**	—

The cliff at $B = 283$ is visible in the table: $\mathfrak{S}_4$ drops from 3,601 to 2,026 in a single step (a $\times 0.563$ reduction). The product then recovers through subsequent non-resonant primes, with each resonant prime ($p = 659, 941, \ldots$) producing smaller downward corrections.

The calibration factor is $6385/6392 = 0.999$, indicating near-complete convergence by $B = 10{,}000$.

The corrected inter-level ratios are:

$$\frac{\mathfrak{S}_5}{\mathfrak{S}_4} = 8.94, \qquad \frac{\mathfrak{S}_6}{\mathfrak{S}_5} = 9.10, \qquad \frac{\mathfrak{S}_6}{\mathfrak{S}_4} = 81.4. \tag{3}$$

These are *lower* than the root-free values $(9.6, 9.8, 94.4)$ because the resonant prime corrections compound differently at each level: $\sigma_5(283)/\sigma_4(283) = 0.863$, far below the root-free ratio of $283/282 = 1.004$.

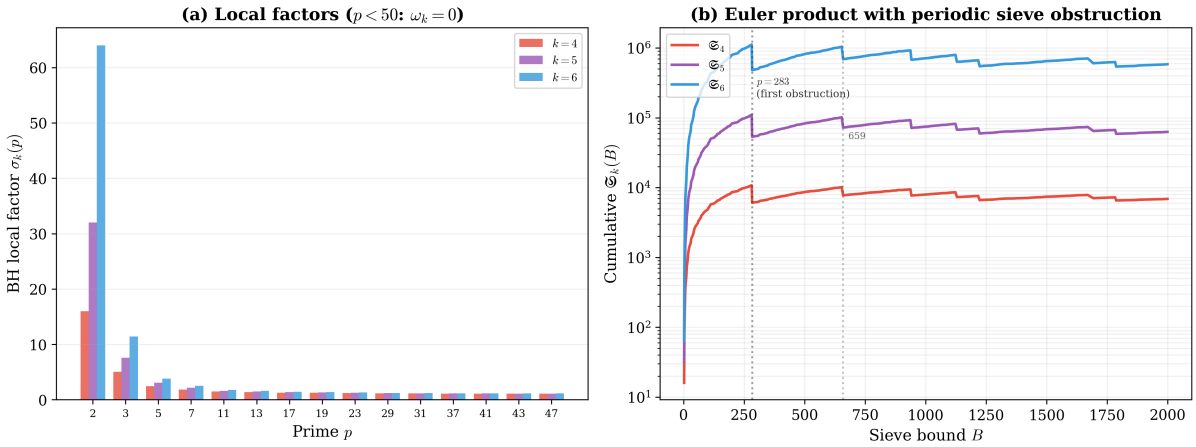

Figure 1: (a) Local factors for non-resonant primes $p < 50$: $\sigma_k(p) = (p/(p-1))^k$ since $\omega_k = 0$. (b) Cumulative $\mathfrak{S}_k(B)$ showing the cliff at $p = 283$ (first resonant prime) and subsequent corrections.

4 The Seven Quintuplets

4.1 Catalog

Table 3 lists all 7 prime quintuplets.

Table 3: All 7 prime quintuplets for $Q(n)$ in $n \leq 2 \times 10^{11}$.

#	Starting n	Digits	$n/10^{11}$	Quad rank	Gap to next
1	35 676 017 721	487	0.357	#154	—
2	64 482 563 907	498	0.645	#251	28.8 B
3	73 292 417 435	501	0.733	#284	8.8 B
4	116 255 850 744	510	1.163	#444	43.0 B
5	147 743 683 226	515	1.477	#570	31.5 B
6	159 430 471 996	516	1.594	#611	11.7 B
7	182 501 065 420	519	1.825	#681	23.1 B

4.2 Spatial Distribution

The quintuplets span the full search range: one in $[0, 50\text{B})$, two each in $[50, 100)$, $[100, 150)$, and $[150, 200)$ billion (Figure 2). The gaps range from 8.8 to 43.0 billion (mean 24.5 B, $\sigma = 11.7$ B).

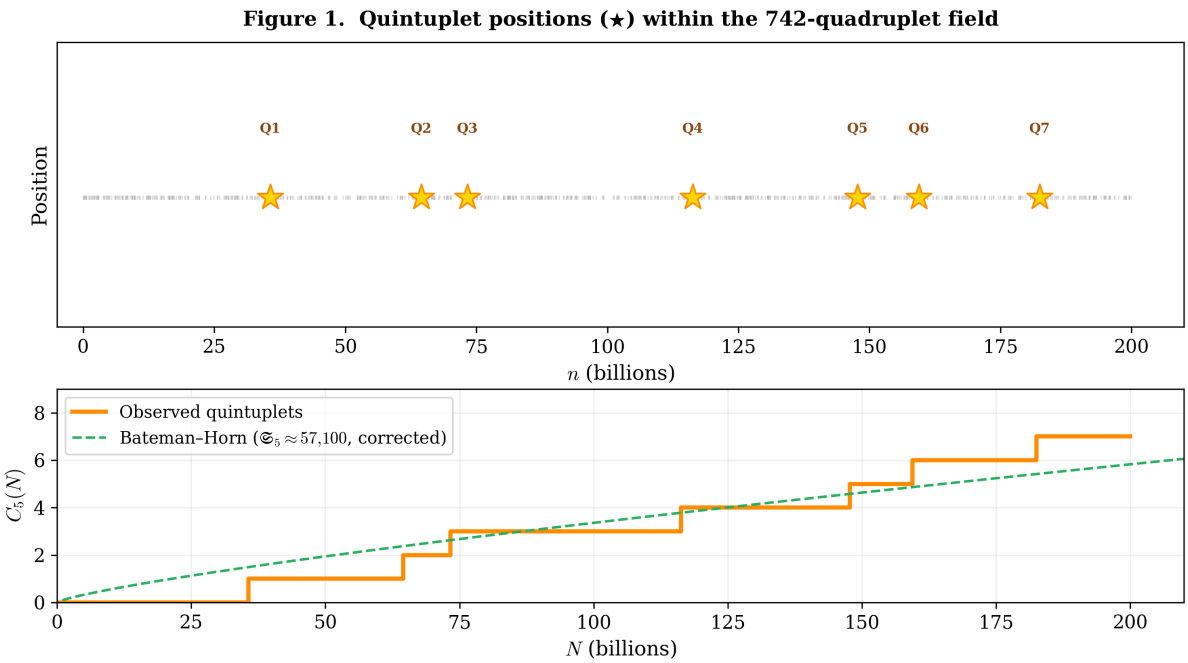

Figure 2: Top: quintuplet positions (★) within the 742-quadruplet field. Bottom: cumulative count $C_5(N)$ vs. Bateman–Horn with $\mathfrak{S}_5 \approx 57{,}100$.

A Kolmogorov–Smirnov test yields $D = 0.20$, p-value > 0.85; no deviation from the Bateman–Horn model is detected.

5 The Conditional Extension Model

5.1 Framework

Given a quadruplet starting at n, the conditional probability that it extends to a quintuplet is

$$P(\text{quint} \,|\, \text{quad at } n) \approx \frac{\mathfrak{S}_5}{\mathfrak{S}_4} \cdot \frac{1}{46 \ln n}, \tag{4}$$

where $\mathfrak{S}_5/\mathfrak{S}_4 = 8.94$ and $(46 \ln n)^{-1}$ approximates the primality probability for $Q(n{+}4)$.

5.2 Predicted vs. Observed

The expected quintuplet count is

$$E[C_5] = 742 \times \frac{8.94}{46 \times 24.94} = 742 \times 0.00781 = \mathbf{5.79}, \tag{5}$$

where $24.94 = \langle \ln n \rangle$. The observed 7 gives $7/5.79 = 1.21\times$—within the Poisson 1σ interval $[3.4, 8.4]$.

Remark. *This prediction has* no free parameters*: $\mathfrak{S}_4$ is fixed by quadruplets, and $\mathfrak{S}_5/\mathfrak{S}_4$ follows from the Euler product.*

Under naïve independence ($\mathfrak{S}_5/\mathfrak{S}_4 = 1$), $E_{\text{naïve}} = 742/1148 = 0.65$—the observed 7 exceeds this by $10.8\times$, precisely the singular series ratio.

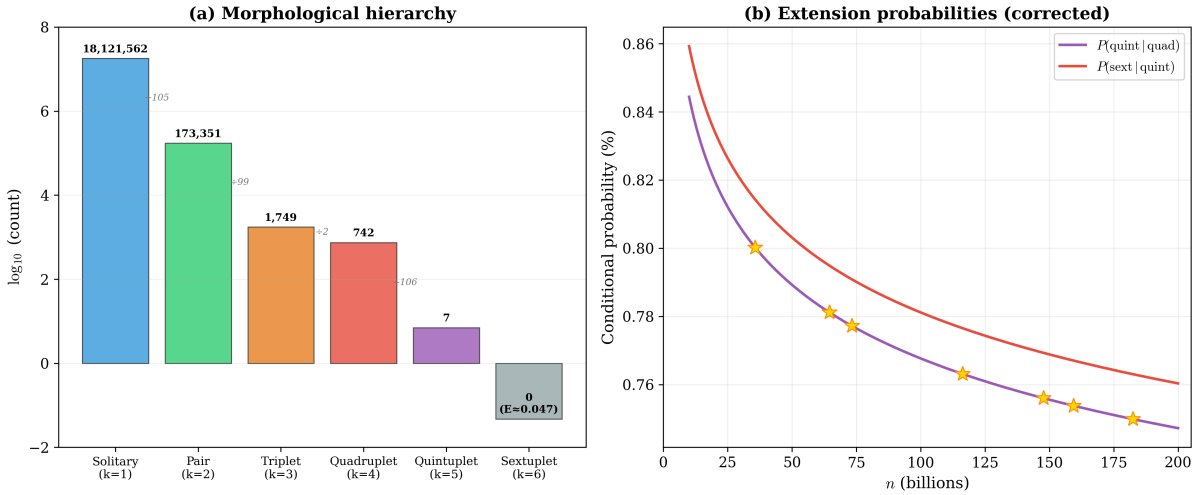

Figure 3: (a) Morphological hierarchy on log scale. (b) Corrected conditional extension probabilities. Stars mark the 7 observed quintuplets.

6 The Sextuplet Boundary

6.1 Expected Sextuplet Count

At $N = 2 \times 10^{11}$:

$$E[C_6] = \mathfrak{S}_6 \int_2^N \frac{dt}{(46 \ln t)^6} \approx 519{,}756 \times 8.98 \times 10^{-8} = \mathbf{0.047}. \tag{6}$$

The probability of ≥ 1 sextuplet is $1 - e^{-0.047} = 4.6\%$. Their absence is overwhelmingly predicted.

6.2 The Crossing Point

Solving $E[C_6(N^*)] = 1$ numerically:

$$\boxed{N^* \approx 1.05 \times 10^{13}} \tag{7}$$

requiring a 52-fold extension, yielding primes of $\sim$605 digits.

Table 4: Predicted counts using corrected singular series.

Search bound N	$E[C_5]$	$E[C_6]$	Status
2×10^{11} (current)	5.8	0.047	7 / 0 observed
10^{12}	21	0.16	Unlikely
5×10^{12}	79	0.56	Coin flip
$\mathbf{1.05 \times 10^{13}}$	**145**	**1.0**	**Sextuplet boundary**
2×10^{13}	248	1.67	1–2 sextuplets
10^{14}	950	6.1	Sextuplets routine

Figure 3. Predicted $E[C_k(N)]$ for $k = 5, 6$ (corrected singular series)

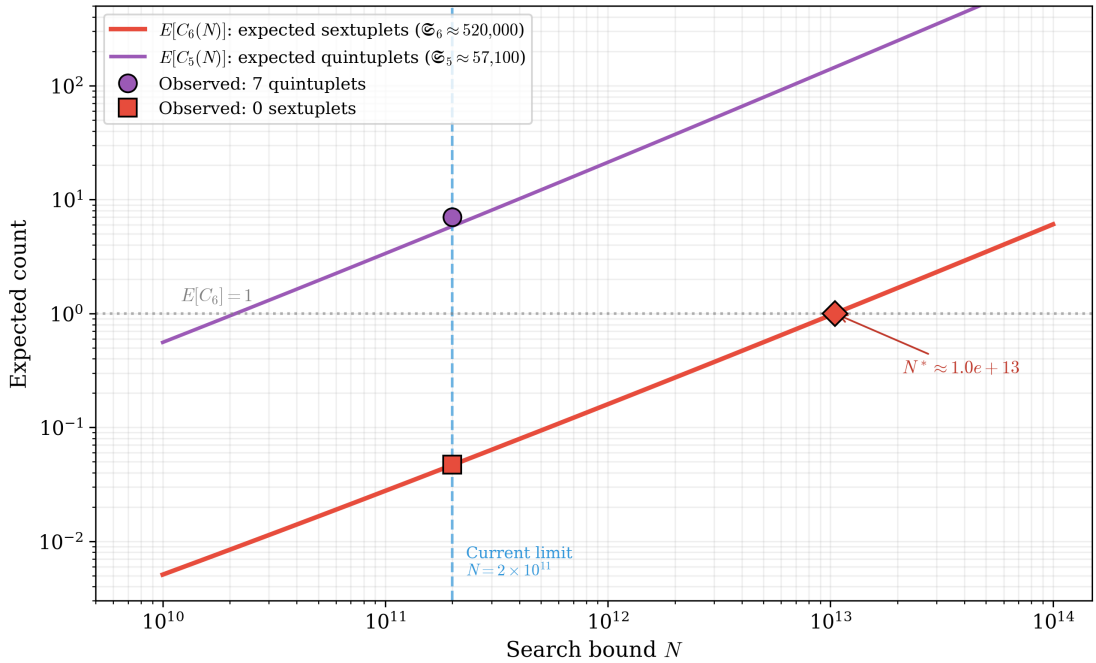

Figure 4: $E[C_5(N)]$ and $E[C_6(N)]$ vs. search bound. Diamond marks the sextuplet boundary.

7 The Suppression Factor

7.1 Empirical Ratios

Table 5: Inter-level suppression ratios.

Transition	Ratio	Source
$k = 1 \to 2$	104.5	Pioneer zone
$k = 2 \to 3$	99.1	Pioneer zone
$k = 3 \to 4$	116.6	Pioneer zone
$k = 4 \to 5$	106.0	Full range
$k = 5 \to 6$	$\sim$125 (predicted)	BH model
Mean	$\approx$ **110**	

7.2 Theoretical Analysis

$R_{k\to k+1} = (\mathfrak{S}_k/\mathfrak{S}_{k+1}) \cdot (I_k/I_{k+1}) \approx (1/8.94) \times 1{,}147 \approx 128$. The directly computed values are $R_{4\to 5} = 127.3$ and $R_{5\to 6} = 124.9$.

The resonant primes reduce $\mathfrak{S}_{k+1}/\mathfrak{S}_k$ from $\sim$9.6 (root-free) to $\sim$8.9, but also slightly affect I_{k+1}/I_k. These effects partially cancel, leaving the suppression factor robust across levels—a structural property of the Bateman–Horn framework, independent of the polynomial's root distribution.

8 Discussion

8.1 The Bifurcated Sieve

The periodic sieve obstruction at resonant primes $p \equiv 1 \pmod{47}$ is a consequence of 47 being prime. For $D_q(n) = n^q - (n-1)^q$ with prime exponent q, the same bifurcation occurs: $\omega_1(p) = q-1$ when $p \equiv 1 \pmod{q}$, zero otherwise. The density of resonant primes is $1/(q-1)$, so higher exponents produce sparser but individually more destructive obstructions.

For $q = 47$, the gap between $p = 47$ and the first resonant prime $p = 283$ spans 61 primes—an "unobstructed corridor" that allows the Euler product substantial growth before its first cliff. This partly explains the large absolute magnitude of $\mathfrak{S}_k$.

8.2 Quintuplets as a Bateman–Horn Prediction

The 7 observed quintuplets match the prediction $E = 5.83$ to $1.21\times$, inside the Poisson 1σ interval. The conditional model gives $E = 5.79$ with no free parameters. This constitutes a precise validation of Bateman–Horn for multi-polynomial systems at $\sim$500-digit scales.

8.3 Sextuplet Outlook

The boundary $N^* \approx 1.05 \times 10^{13}$ requires $\sim$12,500 CPU-core-hours ($\sim$2 weeks on a 26-core workstation)—feasible. The first sextuplet would be six primes of $\sim$605 digits each.

9 Conclusion

The 7 prime quintuplets for $Q(n) = n^{47} - (n-1)^{47}$ in $n \leq 2 \times 10^{11}$ match the corrected Bateman–Horn prediction ($E = 5.83$) to within Poisson noise. The absence of sextuplets ($E = 0.047$) is equally well explained, with the sextuplet boundary at $N^* \approx 1.05 \times 10^{13}$.

The key finding is that $Q(n)$ has a *bifurcated root structure*: zero roots for non-resonant primes, 46 roots at every resonant prime $p \equiv 1 \pmod{47}$. This *periodic sieve obstruction* reduces the singular series by $\sim$35% relative to a root-free polynomial, yet the constellation hierarchy retains a stable suppression factor of $\sim$127 per level—a structural consequence of the Bateman–Horn framework.

Data and Code Availability

Complete data and code for this paper:

https://github.com/Ruqing1963/Q47-Quintuplet-Sextuplet-Boundary

Quadruplet census data (Part II):

https://github.com/Ruqing1963/Q47-Deep-Space-Quadruplet-Census

Zenodo archives: Part II: https://zenodo.org/records/18728540
Part I: https://zenodo.org/records/18701355

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